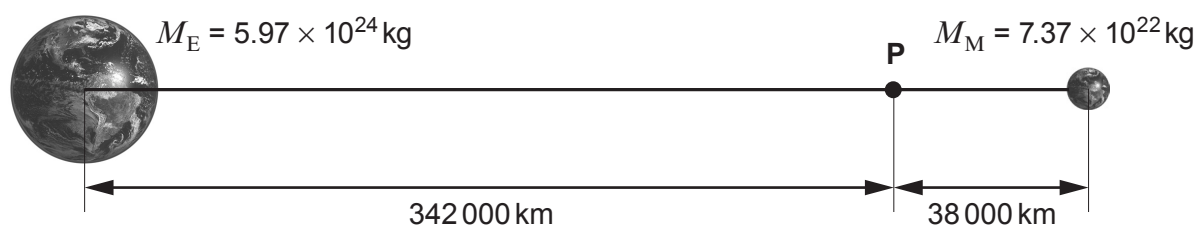


2. (a) Assuming that the Earth is an isolated perfect sphere, draw its gravitational field lines and equipotential surfaces. [3]



- (b) (i) Use the information in the diagram to calculate the gravitational potential at point P. [3]



- (ii) Use the information in the diagram to show that the resultant gravitational field at point P is very small. [2]



- (iii) Myfanwy correctly calculates that the force on a 25 tonne spaceship would be negligible at point **P** and that the force would increase by approximately 0.5 N for every 10 km moved away from point **P** towards the Earth. Dafydd then concludes that the spaceship will perform simple harmonic motion about point **P**. Deduce whether or not Dafydd is correct (**no further calculations are required**). [3]

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